

Formula Sheet for Final Exam

1. Enterprise Value = Equity + Net Debt,
where Net Debt = Debt – Cash
2. (Firm) Free Cash Flow = EBIT $(1 - \tau_c)$ + Depreciation & Other Non-Cash Items – Capital Expenditures – Δ Net Working Capital,
where τ_c is the marginal corporate tax rate.
3. Equity Free Cash Flow = (EBIT – Interest) $\times (1 - \tau_c)$ + Depreciation & Other Non-Cash Items – Capital Expenditures – Δ Net Working Capital + Δ Debt
4. Liquidation or Salvage Value = Sale Price – $(\tau_c \times \text{Capital Gain})$,
where Capital Gain is the difference between Sale Price and Book Value.
5. Growing perpetuity formula: $PV = \frac{CF_1}{r-g}$,
where r is the discount rate, g the growth rate, and CF_1 the cash flow realized one period from now.
6. Growing annuity formula: $PV = \frac{CF_1}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^N \right]$,
where N is the number of periods the annuity is paid.
7. Earnings Growth Rate = $(1 - \text{Payout Rate}) \times \text{Return on New Investment}$
8. CAPM: $r_i = r_f + \beta_i \times MRP$,
where r_f is the risk-free rate, β_i is the beta of security i , and MRP is the market risk premium.
9. (After-tax) Weighted Average Cost of Capital: $r_{wacc} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_c)$,
where E and D refer to Equity and (net) Debt, respectively.

10. Levering formulas assuming that risk of the tax shields = risk of unlevered assets:

$$r_E = r_U + \frac{D}{E} (r_U - r_D) \quad \text{and} \quad r_U = \frac{D}{D+E} r_D + \frac{E}{D+E} r_E,$$

$$\beta_E = \beta_U + \frac{D}{E} (\beta_U - \beta_D) \quad \text{and} \quad \beta_U = \frac{D}{D+E} \beta_D + \frac{E}{D+E} \beta_E,$$

where r_U is the unlevered cost of capital and β_U is the unlevered asset beta.

11. Levering formulas assuming that risk of tax shields = risk of debt:

$$r_E = r_U + \frac{D}{E} (r_U - r_D) (1 - \tau_c) \quad \text{and} \quad r_U = \frac{r_E + (D/E)(1 - \tau_c)r_D}{1 + (D/E)(1 - \tau_c)},$$

$$\beta_E = \beta_U + \frac{D}{E} (\beta_U - \beta_D) (1 - \tau_c) \quad \text{and} \quad \beta_U = \frac{\beta_E + (D/E)(1 - \tau_c)\beta_D}{1 + (D/E)(1 - \tau_c)}.$$